Project Documentation: Car Price Prediction

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1. Project Overview

- **Project Title:** Used Car Price Prediction using Machine Learning
- **Objective:** To build and evaluate machine learning regression models capable of predicting the selling price of used cars based on historical data and vehicle specification.

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2. Problem Statement

Pricing a used car accurately is challenging due to the various factors involved, such as the car's age, mileage, fuel type, and transmission. This project aims to automate the valuation process by training a predictive model on a dataset of past car sales, enabling accurate estimations of a car's Selling_Price.

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3. Dataset Description

- **File:** car data.csv

- **Size:** 301 entries (rows) and 9 features (columns).
- **Target Variable:** Selling_Price (Continuous float value).
- **Independent Variables:**
  - Car_Name: Name of the vehicle.
  - Year: Manufacturing year.
  - Present_Price: Current ex-showroom price.
  - Kms_Driven: Total distance the car has traveled.
  - Fuel_Type: Categorical (Petrol, Diesel, CNG).
  - Seller_Type: Categorical (Dealer, Individual).
  - Transmission: Categorical (Manual, Automatic).
  - Owner: Number of previous owners.

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4. Data Preprocessing & Exploratory Analysis

Initial data inspection confirmed that there were **zero missing values** across all 301 rows, eliminating the need for missing value imputation.

To prepare the data for machine learning algorithms, the categorical text features were manually mapped and encoded into numerical integer values:

- **Fuel_Type:** Encoded as Petrol: 0, Diesel: 1, CNG: 2.
- **Seller_Type:** Encoded as Dealer: 0, Individual: 1.
- **Transmission:** Encoded as Manual: 0, Automatic: 1.

(Note: The Car_Name column was dropped prior to training as it is a high-cardinality text identifier that does not contribute linearly to the models used.)

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5. Model Selection & Training

The processed dataset was split into dependent (Y) and independent (X) variables. The data was then divided into training and testing sets using a **90% training / 10% testing** split (test_size=0.1) with a random_state of 2 for reproducibility.

Two regression algorithms were selected and trained on the data:

- **Linear Regression:** Used as the primary baseline model to establish a linear relationship between the car features and the selling price.
- **Lasso Regression (L1 Regularization):** Used to observe if penalizing absolute weights would improve generalization and handle potential multicollinearity better than standard linear regression.

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6. Evaluation Metrics & Results

The models were evaluated using the **R^2 (R-squared) Error** metric, which indicates the proportion of the variance in the dependent variable that is predictable from the independent variables.

Scatter plots were also generated for both models to visually compare the Actual Prices against the Predicted Prices.

Model 1: Linear Regression Performance

- **Training Data R^2 Score:** 0.8799
- **Testing Data R^2 Score:** 0.8365
- *Observation:* The model performed well on training data but showed a slight drop in accuracy on unseen test data, indicating mild overfitting.

Model 2: Lasso Regression Performance

- **Training Data R^2 Score:** 0.8427
- **Testing Data R^2 Score:** 0.8709
- *Observation:* While the training score was slightly lower than Linear Regression, the

Lasso model generalized significantly better to the unseen test data.

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7. Conclusion

The **Lasso Regression model is the superior model** for this specific dataset. By applying L1 regularization, it successfully prevented overfitting and yielded a higher accuracy (R^2 score of ~87%) on the test data compared to the standard Linear Regression model.